

Academic Statement of Purpose

Program: Computer Science and Engineering M.S.E.

Applicant: Yang Sen (Liam) Lin

*I aspire to **architect trustworthy embodied intelligence** by unifying the semantic reasoning of vision-language models with the provable safety guarantees of control theory.*

In my recent research accepted to **CoRL 2025**, “**See, Point, Fly**,” I identified a fundamental trade-off in robotic autonomy: geometric rules provide reliability but become brittle in unstructured environments; conversely, Generative Vision-Language-Action (VLA) models offer adaptability but often suffer from hallucinations and high computational latency. My objective at the University of Michigan is to resolve this tension. By integrating the formal verification methods of **Prof. Necmiye Ozay** with the uncertainty-aware semantic planning of **Prof. Bernadette Bucher**, I aim to architect “**Trustworthy Embodied Systems**” that constrain the creativity of VLMs with the safety guarantees of control theory.

My journey toward this objective began with a focus on algorithmic decoupling. In my **CoRL 2025** work, I formulated a geometric projection algorithm to ground 2D VLM prompts into 3D metric controls, solving depth ambiguity via pin-hole camera models without external sensors. By framing action prediction as a spatial grounding task rather than end-to-end text generation, I achieved **state-of-the-art success rates (93.9%)** in unstructured environments. However, through rigorous testing, I observed that my “learning-free” geometric priors became brittle when facing erratic, dynamic obstacles where non-linear reasoning is required. To overcome this “**reactivity bottleneck**” caused by VLM inference latency, I realized I must evolve from static heuristics to adaptive, safety-critical policies.

I am particularly drawn to UMich’s research-oriented curriculum, which offers the ideal environment to investigate how **hybrid system controls** can replace static rules. My technical roadmap creates a bridge between probabilistic AI and deterministic control through two strategic academic pillars.

First, I aim to collaborate with **Prof. Necmiye Ozay** to develop “**safety filters**” that use control-theoretic guarantees to prevent VLM hallucination. While my previous work relied on heuristic safety constraints, I intend to leverage coursework such as **EECS 563 (Hybrid Systems Control)** to master correct-by-construction control synthesis. Gaining this theoretical foundation is critical for my goal of formalizing safety constraints that can mathematically bound the output of probabilistic VLM policies, ensuring agents respect physical dynamics even when the VLM makes errors.

Simultaneously, I seek to engage with **Prof. Bernadette Bucher**’s research group to address the generalization limits I encountered in my CoRL work. Her focus on **planning under uncertainty** perfectly complements my background. I plan to apply my **Sim2Real experience** to her ongoing work in vision-language grounding, moving beyond my current zero-shot frameworks to few-shot adaptive systems. To support this, I plan to enroll in advanced topics courses like **EECS 598 (Learning-based Robotics)** to investigate data-driven approaches like visual foresight, allowing me to extend my navigational frameworks into highly dynamic environments.

My research philosophy is further defined by the constraints of the physical layer. While many researchers focus solely on algorithmic performance, my internship at **Wolley Inc.** grounded me in the realities of

hardware deployment. There, I engineered C-based diagnostic tools to profile **CXL memory bandwidth**, gaining the low-level intuition needed to optimize I/O efficiency for physical AI. I view this not just as hardware engineering, but as a critical step in **reducing the inference latency of foundation models**. At UMich, I hope to take **EECS 598 (Systems for Generative AI)** to explore how to optimize these heavy models for edge deployment. This unique combination of formal methods and systems engineering is essential for building real-time embodied AI that is deployable in the real world.

My academic preparation is defined by the convergence of **Mechanical Engineering and Computer Science** at NYCU. While ME grounded me in the physical first principles of system dynamics, my dual focus in CS equipped me with the algorithmic reasoning of modern AI. I demonstrated the ability to synthesize these fields by securing an **A+ in Graduate Computer Vision** and applying Projective Geometry to derive non-linear unprojection laws in my CoRL paper. This practical application prepares me for the rigorous theoretical training of **ROB 501 (Mathematics for Robotics)**, where I look forward to solidifying my foundations in non-linear optimization and set theory.

Beyond the laboratory, I carry an “optimization mindset” from code to community. As the **GDSC Chapter Lead**, I directed 24 members to bridge the gap between academic theory and industry practice, organizing workshops that impacted over 40 student developers. I aim to bring this spirit of collaborative innovation to UMich, specifically by **facilitating dialogue between the CSE and Robotics cohorts**. I thrive in interdisciplinary environments and look forward to organizing study groups or hackathons that encourage peers to test algorithms on the diverse hardware platforms available at the **Robotics Institute**.

My long-term goal is to lead the R&D of **Ubiquitous Physical AI**. The UMich MSE in CSE is the pivotal step that transforms me from a robotics engineer who builds working prototypes into a **Computational Systems Architect**. The program’s structured training in formal verification and system optimization will provide me with the rigorous toolkit necessary to define the safety guarantees required to deploy autonomous agents at a global scale.